

E-NPU: A 34~126nJ/Class Event-Driven Adaptive Neural SoC with Signal-Dynamics-Aware Feature Clustering and Multi-model In-Memory Inference/Training for Personalized Medical Wearables

Fengshi Tian^{1,2,*}, Jinbo Chen^{3,*}, Kunming Shao^{1,2,*}, Zilu Liu¹, Jiakun Zheng^{1,2}, Hui Wu³, Chaoming Fang³, Xiaomeng Wang^{1,2}, Ziyang Shen³, Pingcheng Dong^{1,2}, Yuan Yao¹, Xuliang Wang⁴, Jie Yang³, Mohamad Sawan³, Chi-Ying Tsui^{1,2}, Kwang-Ting Cheng^{1,2}

* Equally-Credited-Authors (EACs)

¹HKUST, Hong Kong SAR; ²AI Chip Center for Emerging Smart Systems, Hong Kong SAR; ³Westlake University, China; ⁴Tsinghua University, China

Biomedical signal processing systems-on-chip (SoC) have shown significant promise in human-machine interfaces and closed-loop neuromodulation applications [1-5]. Future personalized medical wearables are anticipated to provide long-term monitoring and accurate interpretation while maintaining minimal energy consumption [6-8]. Lightweight neural networks, such as binary, spiking, and convolutional neural networks (BNN/SNN/CNN), are promising for neural signal analysis, with neural architecture search (NAS) techniques enabling the deployment of task-specific models optimized for accuracy and efficiency [9-12]. However, developing an accurate, ultra-energy-efficient medical SoC that seamlessly integrates signal feature extraction and processing with multi-model reconfigurability remains challenging. Furthermore, inter-patient variance adversely affects the accuracy of biosignal classifications in real-world scenarios, thereby limiting the reliability of these devices [13,14].

To address these challenges, we propose an event-driven neural SoC (E-NPU) with three key innovations to enhance energy efficiency and accuracy in personalized medical wearables (Figure 1). 1) **Self-Adaptive Processing Architecture**: This architecture includes a sparse signal-dynamics-aware analog-to-feature converter and event-driven two-stage processing engines, reducing average energy consumption by over 1.7x in biosignal feature extraction and classification by replacing conventional always-on processing. 2) **Multi-Model Multi-Precision CIM Cores**: We propose reconfigurable computing-in-memory (CIM) cores that support task-aware NAS-decided BNN/SNN/CNN classification. This approach addresses the memory wall issue and demonstrates prior accuracy (95.6%~99.6%) and runtime energy efficiency (34~126nJ/Class) for multi-task scaling. 3) **Direct Error Feedback On-Chip Training**: To counteract accuracy degradation from inter-patient variance, we design a CIM engine supporting floating-point precision for direct error feedback local training. This engine facilitates long-term patient-specific model updates using layer-wise local error propagation, significantly reducing on-chip storage needs compared to traditional backpropagation. Additionally, we propose inter-core operation pipelines to decrease overall inference/training latency by 3.03x and 2.49x, respectively, and external memory access (EMA) by 3.41x.

Figure 2 illustrates the overall architecture of the proposed E-NPU SoC. It comprises a 6-channel signal-dynamics-aware analog-to-feature converter (SAFC), a self-adaptive event-driven 2-stage operator (ED-Op), a heterogeneous CIM inference/training engine with 4 event-driven CIM (ED-CIM) cores and 4 local-training CIM (LT-CIM) cores, a system data router for managing inter-block data transfer, cache memory (MEM) for storing intermediate data and neural network (NN) parameters, and a SoC controller (CTRL) for instruction decoding, single-instruction-multiple-data (SIMD) operations, and power/clock gating configurations. Analog inputs are received from external multi-channel analog front ends (AFE) and directed to the SAFC for sparse feature generation. Traditional methods [1,4,5], which use ADCs to digitize analog inputs followed by FIR/IIR for spectral analysis, are computationally intensive and fail to capture the temporal dynamics of time-series biomedical signals. Our SAFC overcomes these limitations by detecting dynamic amplitude changes in input signals using a level-cross detector (LCD), which generates sparse spike trains indicating signal increases (UP) and decreases (DN). These spike trains are processed by the event-to-feature generator (EFG), which uses them as indicators for the +1/-1 of a predefined index. The EFG encodes the index into a binary vector during each cycle, forming part of the output binary feature map (BFM). The BFMs are subsequently down-sampled in the compressor for backend classification. The LCD's sensing resolution is self-adaptive, catering to different signal types (e.g., EEG, EMG, ECG) with an event-triggered feedback loop that automatically generates appropriate detection windows for various tasks.

The binary feature maps (BFMs) generated by the SAFC are buffered in MEM and sent to the ED-Op for adaptive processing. Figure 3 details the key circuits involved. In user-defined mode, instructions specify the neural network modality (BNN/SNN/CNN) and data precision, with configuration bitmaps dispatched to the four ED-CIM cores via the SIMD core. In self-adaptation mode, BFMs are directed to a trainable feature cluster (FC) for abnormality detection, which is the 1st stage classifier. The FC comprises a 2D convolution unit, a 2D max-pooling unit, and a

similarity comparator utilizing difference hash and Hamming distance. Input BFMs are compared against a reference BFM stored in MEM. If the similarity falls below a certain threshold, an abnormality is detected, triggering a wake-up signal to activate the SIMD core and ED-CIM cores for neural network classification. To address inter-patient variance, the FC undergoes training upon detecting abnormalities. If the neural network classification result differs from the FC (normal vs. abnormal), the reference BFM is updated to incorporate patient-specific data. Figure 3 also showcases the detailed ED-CIM circuits. Each core contains 64 parallel integration columns equipped with 64x2-bit ping-pong (PP) SRAM cells connected to a column-wise 4-2 compressor adder. Column outputs are fed into a reconfigurable output fusion module with precision scaling and neural activation units. PP cells on the same row share a common input driver, and the two 6T SRAM cells within each PP cell operate in a ping-pong pipeline manner, effectively doubling operation throughput. The mapping of BNN/SNN/CNN onto ED-CIM is also illustrated. BNNs feature multi-bit inputs and binary weights, with inputs serialized from least significant bit (LSB) to most significant bit (MSB). SNNs have binary inputs/outputs across multiple time steps, utilizing a time-step-first mapping method to fully exploit CIM's parallel capabilities. For SNNs and CNNs, which employ multi-bit weights, column outputs are grouped and scaled using shift-add (SA) units. CNN inputs are processed in CIM serially from LSB to MSB, similar to BNNs. To minimize latency overhead from reconfigurability and enhance resource utilization, we developed a two-step output fusion module with built-in pipelines. The first step involves precision scaling on the SA array, with SA units hierarchically cascaded and bypass paths activated for lower bit precision. The second step involves neural network activation with three distinct activation functions for BNN/SNN/CNN. We divided the activation function into three sub-steps, reusing the most power-intensive arithmetic logic units. Serial-in-out and parallel-in-out for CNN/BNN and SNN are achieved by controlling output timing with local counters.

Figure 4 illustrates the direct feedback local training (LT) dataflow and its implementation on a CIM-based training engine. Local training is executed layer by layer, which unlocks parallelism by eliminating cross-layer data dependencies, making it ideal for on-chip, end-to-end neural network training with patient-specific data [15-18]. The LT process for each layer involves three steps: 1) **Precision Conversion**: Transforming data from fixed-point integer (INT) to floating-point (FP) format. 2) **Local Error Prediction (LEP)**: Generating layer-specific local errors using average pooling (AP) units and a fully-connected layer classifier (FCC) that aligns with the neural network's final layer. 3) **Direct Feedback Update (DFU)**: Replacing the layer with new weights, involving pseudo-random number generation (PRNG), direct gradient generation (DGG), and weight update (WU). DGG processes errors from LEP and a random matrix from PRNG to transmit weight gradients to WU. The updated weights are then converted back to INT to refresh the trained layer. The entire LT flow is mapped across four LT-CIM cores, which interact with the SIMD core, ED-CIM cores, and LT-support module (LTSM). The SRAM arrays in LT-CIM cores share the same ping-pong structure as those in ED-CIM cores. During training, the FCC in LEP and DGG in DFU are executed by LT-CIM cores, while other steps are managed by units in LTSM. An intra-core pipeline scheme, based on the ping-pong structure, is employed on LT-CIM to reuse output peripheral circuits and minimize core-to-memory intermediate data movements. To further reduce end-to-end processing latency for diverse medical tasks with compact neural network structures, inter-core operation pipelines are developed for both inference (IP) and training (TP). In IP, activation transfers are pipelined in a layer-fusion pattern, enhancing ED-CIM utilization. In TP, pipelines are established between LTSM and LT-CIM cores, decreasing external memory access (EMA).

The design is fabricated using a 40nm CMOS process. Figure 5 presents measurement results from extensive benchmarks and testing scenarios, featuring a neural signal analysis example utilizing our event-driven operation. Accuracy results across multiple benchmarks highlight the effectiveness of the proposed on-chip training scheme for both two-stage classifiers, with 60% of the data allocated for pre-training and 30% for re-training. Evaluated with a modified LeNet, inter-core acceleration results show that the inference pipeline (IP) reduces latency by 3.03x and enhances ED-CIM utilization by 1.86x, while the training pipeline (TP) reduces training latency by 2.49x and external memory access (EMA) by 3.41x. The real-time energy efficiencies of the feature cluster (FC) and the multi-NN multi-precision CIM engine are assessed on four public datasets: EEG motor imagery [19], EEG seizure detection [20], SeNc EMG gesture [21], and MIT-BIH cardiac arrhythmia [22]. Energy consumption for multi-precision in-memory BNN/SNN/CNN operations is measured at 50MHz with a 0.9V supply. The proposed ED-Op scheme achieves over 1.7x energy reduction across the four tasks. Additionally, the average system accuracy improves with ED-Op, as the combination of two-stage classifiers reduces the overall miss rate. Figure 6 compares our work's performance with state-of-the-art designs, demonstrating that the E-NPU provides superior accuracy and energy efficiency with multi-task flexibility. Figure 7 displays the chip micrograph.

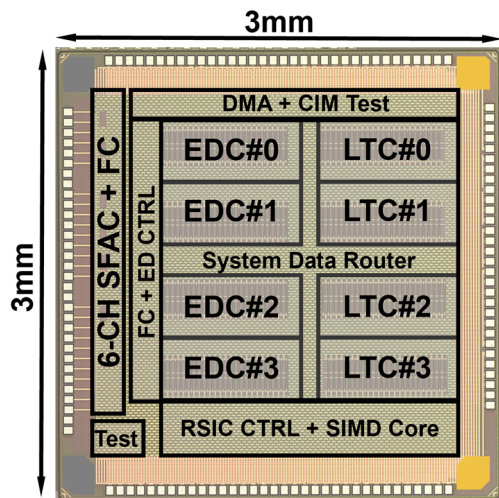


Fig. 7 Chip micrograph.

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